# MISCELLANEOUS NOTES IN PHYSICS

## And other stuff

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# 1 Different Approaches to Quantum Dynamics

The two approaches, or pictures as we call them, serve as two different ways of thinking about what changes in our system with time and what doesn't. This is has more to do with the framework we're setting up and (I think) less with things like conserved quantities and such. The results we calculate won't change and the physics remains the same. But the difference lies in the way of dealing with various objects of the theory. This can make a problem easier to conceptualize and deal with - so there are benefits to using one approach over the other.

## 1.1 Schrödinger Picture (SP)

This is the approach usually adopted for teaching a first quantum mechanics course. Under this, the state kets are assumed to be the dynamical objects that change with time.

So, to have our state ket evolve over time, we'd need the time-evolution operator $\mathcal{U}$. With the observable $H$ fixed in time and $\mathcal{U}_t = e^{-\frac{iHt}{\hbar}}$, the state at time $t$ is given by,

$$|\alpha; t\rangle = \mathcal{U}_t \, |\alpha; t_0\rangle \tag{1}$$

Let the expectation value of an observable at time $t = 0$ be,

$$\langle A \rangle_0 = \langle \beta; 0| \, A \, |\alpha; 0\rangle \tag{2}$$

the expectation value at a later time $t$ is,

$$\langle A \rangle_t = \langle \beta; t| \, A \, |\alpha; t\rangle = (\langle \beta; 0| \, \mathcal{U}_t^\dagger) \; A \; (\mathcal{U}_t \, |\alpha; 0\rangle) \tag{3}$$

## 1.2 Heisenberg Picture (HP)

This lies closer in spirit to classical mechanics. There are no state kets there, but what changes with time are observables (such as position $x$ or angular momentum $L$). So, we will shift the time dependence onto the operator (observable) by sandwiching it between the time-evolution operators. With $t$ as the single parameter, $\mathcal{U}_t$ and $\mathcal{U}_t^\dagger$ represent two families of operators, or rather two operator valued functions. Thus,

$$A(t) = \mathcal{U}_t^\dagger A \mathcal{U}_t = \mathcal{U}^\dagger(t) \cdot A \cdot \mathcal{U}(t) \tag{4}$$

We can write $A(t) = A_H(t)$ since time-dependent operators appear in the Heisenberg picture, while we write $A = A_S$ since operators are fixed in the Schrödinger picture.

At $t = 0$, the two operators coincide: $A_H(0) = A_S$. This is also true for state kets. However, once clocks start running, the state kets freeze in the Heisenberg picture and the operators freeze in the Schrödinger picture.

Again, the expectation values are the same irrespective of the picture we choose.

### 1.2.1 Heisenberg equation of motion

The fundamental equation of motion in this picture is,

$$\frac{dA_H(t)}{dt} = \frac{1}{i\hbar}[A_H(t), H_H] + \left(\frac{\partial A_S}{\partial t}\right)_H \tag{5}$$

Taking expectation values of both sides of the equation (and remembering that they are equivalent for either picture so we can omit the subscripts) automatically gives us Ehrenfest's theorem,

$$\frac{d\langle A\rangle}{dt} = \frac{1}{i\hbar}\langle[A, H]\rangle + \left\langle\frac{\partial A}{\partial t}\right\rangle \tag{6}$$

## 1.3 Evolution of base kets

It is a common misconception that this whole business of changing states and operators transfers over directly to describe how base kets for our Hilbert space behave. That is, base kets should change in time because states do so in SP while they should remain fixed in HP because states also stay fixed. We emphasize that this is incorrect.

By definition, the base kets of our system corresponding to a particular observable satisfy $A|a\rangle = a|a\rangle$ (they are eigenkets of $A$). Since the observables don't change with time, this equation shows that neither do the base kets.

But in HP, the operators do change with time. So, at time $t$,

$$A_H(t) = \mathcal{U}^\dagger(t)\ A_H(0)\ \mathcal{U}(t) = \mathcal{U}^\dagger(t)\ A_S\ \mathcal{U}(t) \tag{7}$$

So,

$$\mathcal{U}^\dagger(t)\ A_H(0)\ \mathcal{U}(t)\ \mathcal{U}^\dagger(t)\,|a\rangle = a\,\mathcal{U}^\dagger(t)\,|a\rangle \tag{8}$$

$$A_H(t)(\mathcal{U}^\dagger(t)\,|a\rangle) = a(\mathcal{U}^\dagger(t)\,|a\rangle) \tag{9}$$

We then impose the requirement that our base kets be eigenkets of the observable. Th equation shows that at each instant of time, the kets $\mathcal{U}^\dagger(t)\,|a\rangle$ are eigenkets of $A_H(t)$ and so, the base kets must be changing in time ($|a;t\rangle$) to always satisfy this eigenvalue equation.

But this means our base kets rotate in the direction opposite to the how the state kets rotate in the Schrödinger picture (because the $\mathcal{U}$ is changed to $\mathcal{U}^\dagger$, which is the adjoint).

# 2 Perturbation theory

Say you're studying a quantum system that's similar to one you've studied before. The Hamiltonian of this new system doesn't differ much from that of the older system - just an additional new term that's small, in some sense, in its effects [1]. You can imagine introducing this perturbation with a strength factor - although the form of the perturbation is fixed (maybe you're turning on an external magnetic field or introducing a new potential function $V$), this strength factor parameterizes the magnitude of your introduced change. That is, a number $\lambda$ that bridges the gap between no perturbation (for $\lambda = 0$) and complete perturbation (for $\lambda = 1$).

Then, the essence of perturbation theory is that small changes in your Hamiltonian should introduce small changes in your system in a continuous way (of which we keep track through $\lambda$). If your original Hamiltonian had some eigenstates $|n\rangle$ with eigenvalues $E_n$, then the new eigenstates should be close to the original ones (this can be made precise using norms since we're in a Hilbert space) and the energy values should also shift up or down in a continuous manner by small amounts (for small enough perturbations, that is).

There are three situations we will consider here. The first two concern time-independent Hamiltonians, but with non-degenerate and degenerate energy levels respectively. For the final situation, we look at time-dependent Hamiltonians.

## 2.1 Time-independent, non-degenerate

We assume that no correction (of any order) has a component along $\left\langle n^{(0)}\right|$ [2]. Wikipedia calls this an intermediate normalization (why so, I am yet to find out).

We can calculate the l-th order energy correction to be,

$$E_n^{(l)} = \left\langle n^{(0)} \middle| \delta H \middle| n^{(l-1)} \right\rangle$$

A consequence of this fact is that the first order energy correction after a perturbation $\delta H$ of the Hamiltonian is precisely the expectation value of this perturbation on the unperturbed state $\left|n^{(0)}\right\rangle$. That is,

$$E_n^{(1)} = \left\langle n^{(0)} \middle| \delta H \middle| n^{(0)} \right\rangle = \langle \delta H \rangle$$

We can also calculate the l-th order state correction to be,

$$\left|n^{(l)}\right\rangle = \sum_{k \neq n}^{\infty} \left|k^{(0)}\right\rangle \left\langle k^{(0)} \middle| n^{(l)} \right\rangle \tag{10}$$

$$\left|n^{(l)}\right\rangle = \sum_{k \neq n} \left|k^{(0)}\right\rangle \left[ -\frac{\left\langle k^{(0)} \middle| \delta H \middle| n^{(l-1)} \right\rangle}{(E_k^{(0)} - E_n^{(0)})} + \frac{1}{(E_k^{(0)} - E_n^{(0)})} \sum_{j=1}^{l} \left\langle n^{(0)} \middle| \delta H \middle| n^{(j-1)} \right\rangle \left\langle k^{(0)} \middle| n^{(l-j)} \right\rangle \right] \tag{11}$$

[1] I'll get to what I mean by this in a bit.

[2] For a proof, confer Chapter 1 of Barton Zweibach's QM - III notes

This approach works only when we have non-degenerate (unperturbed) energy states - if there's a degenerate energy state ($k \neq n$ but $E_k^{(0)} = E_n^{(0)}$), the denominator for at least one term (in the outermost sum over $k$) blows up and we can't express the state correction as a well-defined sum.

Now, we discuss what it means for the perturbation to be "small", such that our assumption of being able to expand out the new energy eigenstates and eigenvalues as a power series in the parameter $\lambda$ is valid. A way to quantify the size of the perturbation would be the norm of the operator $\delta H$. Now, what limit do we place on this number (which we can theoretically calculate) so that our approximation is reliable? One possible idea would be that the norm should be smaller than the entries/norm or energy eigenvalues of the original Hamiltonian. However, it turns out that it is not the absolute magnitude of the energies (of the original Hamiltonian) that ascertains the validity of this perturbative expansion, but the difference between the energies.

If we're considering the perturbation of the n-th energy level, it matters how far the adjacent energy levels lie. If they're closer together, we would (at least intuitively) expect more mixing from them, giving us bigger energy and state corrections. Since degeneracy becomes a problem, it would also make sense that it is the difference in energies that dictates the shifts of levels after perturbation (because then energy differences are equal to 0, and there may not be any non-zero interval for the $\lambda$ parameter where the expansion formally works).

We also get to see another interesting result, that makes for our first sidebar.

**Sidebar 2.1.** Asymptotic expansions (Eg - anharmonic oscillator with quartic perturbation)

At this stage, you might be wondering if the energy levels of our system cross each other at some stage of the perturbation. Since we can make a one to one association with the unperturbed and perturbed energy states, is it ever the case that the hierarchy of states gets flipped and a lower level jumps above an adjacent upper level? This seems to be a complicated issue which I'd like to fully understand (hopefully) soon, but there is something I'd like to add at this point.

**Sidebar 2.2.** Level repulsion

## 2.2 Time-independent, degenerate

As stated before, the approach adopted to determine the new eigenstates and energy eigenvalues of the perturbed Hamiltonian won't really work. From a mathematical viewpoint, the issue arises due to the denominator of the sum in Eq. 2 from Section 1.1. If there is a degeneracy, the sum would blow up and the correction would be ill-defined.

Let the perturbed Hamiltonian be,

$$H(\lambda) = H^{(0)} + \lambda \delta H \tag{12}$$

In the basis of its eigenstates, let $H$ be the diagonal matrix

$$H^{(0)} = \mathrm{diag}\{E_1^{(0)}, E_2^{(0)}, \dots, E_n^{(0)}, E_n^{(0)}, \dots, E_n^{(0)}, E_{n+1}^{(0)}, \dots\} \tag{13}$$

where $E_n^{(0)}$ repeats $N$ times. This means $H^{(0)}$ has a degenerate $N$ dimensional subspace $V_N$ for eigenvalue $E_n^{(0)}$. Our analysis assumes that this is the only energy level that exhibits degeneracy.

Choose an orthonormal basis for this subspace: $\left|n^{(0)}; 1\right\rangle, \left|n^{(0)}; 2\right\rangle, \dots, \left|n^{(0)}; N\right\rangle$.

The success of this method depends on using the right basis for $V_N$. These basis states are the ones that evolve continuously with the application of the perturbation. Otherwise, the wrong basis states (well, wrong in this context) would seem to discontinuously jump to the appropriate basis states as soon as $\lambda$ is greater than 0. This discontinuity obviously disagrees with the smooth transition that perturbation theory posits.

We write the Hilbert space as a direct sum of $V_N$ and the subspace $V$ of states not in $V_N$:

$$\mathcal{H} = V_N \oplus V \tag{14}$$

Also choose an orthonormal basis for the subspace $V$, namely, $\left|k^{(0)}\right\rangle$ for $k \neq n$. The appropriate orthonormality inner product relations follow.

Pick $k \in \{1, \dots, N\}$. Assume that after the application of the perturbation, the state and its energy evolves as,

$$|n; k\rangle_\lambda = \left|n^{(0)}; k\right\rangle + \lambda \left|n^{(1)}; k\right\rangle + \lambda^2 \left|n^{(2)}; k\right\rangle + \mathcal{O}(\lambda^3) \tag{15}$$

$$E_{n,k}(\lambda) = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \mathcal{O}(\lambda^3) \tag{16}$$

We assert that the perturbed state satisfies,

$$H(\lambda) |n; k\rangle_\lambda = E_{n,k}(\lambda) |n; k\rangle_\lambda \tag{17}$$

This helps us get vector equations for different orders of $\lambda$. Applying bras for different basis states (from $V_N$ and $V$) allows us to extract information about corrections energy and overall states.

Assuming that the degeneracy is lifted at first order - that is, all degenerate states which had energy $E_n^{(0)}$ under the unperturbed Hamiltonian split into distinct energy levels after

perturbation. Then,

$$E_{n,k}(\lambda) = E_n^{(0)} + \lambda \delta H_{nk,nk} - \lambda^2 \sum_{p \neq n} \frac{|\delta H_{p,nk}|^2}{E_p^{(0)} - E_n^{(0)}} + \mathcal{O}(\lambda^3) \tag{18}$$

$$|n;k\rangle_\lambda = \left|n^{(0)};k\right\rangle - \lambda \left( \sum_{p \neq n} \frac{\delta H_{p,nk}}{E_p^{(0)} - E_n^{(0)}} \left|p^{(0)}\right\rangle + \sum_{l \neq k}^{l \leq N} \frac{\left|n^{(0)};l\right\rangle}{E_{n,k}^{(1)} - E_{n,l}^{(1)}} \sum_{p \neq n} \frac{\delta H_{nl,p} \delta H_{p,nk}}{E_p^{(0)} - E_n^{(0)}} \right) + \mathcal{O}(\lambda^2) \tag{19}$$

Of course, it will not always be the case that all initially degenerate levels perfectly split into distinct ones when $\lambda > 0$ (this is only when the first order energy corrections are distinct for all $\left|n^{(0)};k\right\rangle$). What may happen is that some levels split away from others, while a subset keeps to be degenerate at the first order, perhaps splitting into distinct levels at a higher order.

This means we will not be able to figure out the appropriate basis for $V_N$ at the first order, and would need to do more work (up to at least the second order) to determine the optimal basis and thus split the levels. We now show what this would look like. For simplicity however, we consider the case where no splitting occurs at the first order, but all states separate at second order.

< insert stuff about lifting degeneracy at second order here >

## 2.3 Time-dependent

The approach employed for time-dependent perturbation theory is quite different from that for the time-independent case. In fact, the latter rests on a critical assumption that there exists a fixed stationary energy states for our system (that solve the Schrödinger equation), but this isn't necessarily true in the present situation.

The reason is that the existence of energy eigenstates followed form the assumption that the wave function has separable spatial and temporal parts. This is valid for a fixed Hamiltonian but not so much for a time-dependent Hamiltonian.

Our initial objective was to find these energy eigenstates because they form a basis for the state space. This is useful to understand time evolution of any configuration of our system - we could express the current state as a linear combination of these basis states, study their time evolution separately, and the linearity of the SE would allow us to connect all separate terms together to get the time evolved state.

But since there's no guarantee that we will have such eigenstates in the time-dependent $H$ case, we might as well stick to just solving the SE directly. So yes, we're going to have to do this the more direct way. All is not lost however - we can still set up useful machinery via this method. It will have to do with transition probabilities - the probability of finding our system in a particular state after time-evolution under the perturbation (assuming it started from a specific state). This is actually pretty useful, since this is related to ideas

behind quantum mechanical scattering.

Before we start, perhaps it will be beneficial to understand what the interaction picture is.

### 2.3.1 Interaction Picture (IP)

I've previously described the Schrödinger picture and Heisenberg picture briefly. The interaction picture lies somewhere in the middle.

When your Hamiltonian has explicit time dependence, this interpretation helps nicely split that dependence between the operators and states. Assume that your Hamiltonian can be divided into two parts, one fixed and the other evolving in time.

$$H = H^{(0)} + \delta H(t) \tag{20}$$

So, we're treating the time-dependent part as our perturbation (the constraint is that this time dependence has to, again, be small in some sense).

The clean divide between time dependence in the interaction picture is achieved by attributing to the fixed Hamiltonian the Heisenberg picture, and to the time-dependent perturbation an evolving state under the Schrödinger picture. It's almost as if your fixed Hamiltonian is evolving in time in the background, while in the foreground the SE captures the evolution of the state with the (slightly modified) perturbation. In the end, we stick all these moving parts together and reap the profits of this elegant picture [3].

With the fixed Hamiltonian, we can define the time evolution-like operator $\mathcal{U} = e^{-iH^{(0)}t/\hbar}$. We also define a new state,

$$\left|\widetilde{\Psi}(x,t)\right\rangle = \mathcal{U}^\dagger \left|\Psi(x,t)\right\rangle \tag{21}$$

Then this state satisfies the SE,

$$i\hbar\frac{\partial}{\partial t}\left|\widetilde{\Psi}(x,t)\right\rangle = \widetilde{\delta H}(t)\left|\widetilde{\Psi}(x,t)\right\rangle \tag{22}$$

where,

$$\widetilde{\delta H}(t) = \mathcal{U}^\dagger \delta H(t)\mathcal{U} \tag{23}$$

Now, a slick formula to compute expressions of the form $\mathcal{U}^\dagger H\mathcal{U}$.

**Sidebar 2.3.** The Baker–Campbell–Hausdorff formula

Since the fixed Hamiltonian $H^{(0)}$ does have energy eigenstates that form a basis for our

[3] Note that this is called the interaction "picture" because it's all that is - a way to think about the math, which presumably is equivalent.

Hilbert space, we can stipulate,

$$\left|\widetilde{\Psi}(x,t)\right\rangle = \sum_n c_n(t)\left|n\right\rangle \tag{24}$$

Using this expression in Eq. 13, we can derive the following system of first order ODEs for the time-dependent coefficients of the basis states,

$$i\hbar\dot{c}_m(t) = \sum_n \widetilde{\delta H}_{mn}(t)c_n(t) \tag{25}$$

where $\widetilde{\delta H}_{mn}(t) = \langle m|\,\widetilde{\delta H}(t)\,|n\rangle$. Calling $\frac{E_m - E_n}{\hbar} = \omega_{mn}$, we can equivalently say,

$$\widetilde{\delta H}_{mn}(t) = e^{i\omega_{mn}t}\delta H_{mn}(t) \tag{26}$$

which implies,

$$i\hbar\dot{c}_m(t) = \sum_n e^{i\omega_{mn}t}\delta H_{mn}(t)c_n(t) \tag{27}$$

Why is this useful? Well, solving this (infinite) system of coupled differential equations would give us the time varying coefficients of the interaction picture wave function - a solution of the SE, by construction. And with how we've defined this state in Eq. 12,

$$\left|\Psi(x,t)\right\rangle = \mathcal{U}\left|\widetilde{\Psi}(x,t)\right\rangle = e^{-\frac{iH^{(0)}t}{\hbar}}\left|\widetilde{\Psi}(x,t)\right\rangle = \sum_n c_n(t)e^{-\frac{iE_nt}{\hbar}}\left|n\right\rangle \tag{28}$$

we've figured out how our state evolves over time under the effects of the fixed Hamiltonian and the time dependent perturbation (in theory, at least).

### 2.3.2 Perturbative solution

In practice, solving the system of ODEs for the coefficients of the IP state can be difficult. Now that we've developed a robust method of approximating a solution, we can go ahead and solve the interaction picture SE.

Setting up the assumed form of the solution,

$$\left|\widetilde{\Psi}(x,t)\right\rangle = \left|\widetilde{\Psi}^{(0)}(x,t)\right\rangle + \lambda\left|\widetilde{\Psi}^{(1)}(x,t)\right\rangle + \lambda^2\left|\widetilde{\Psi}^{(2)}(x,t)\right\rangle + \mathcal{O}(\lambda^3) \tag{29}$$

Inserting this into the IP Schrödinger equation, we again a vector equation with terms appearing alongside different powers of $\lambda$. Setting all these equal to 0, we get a system of equations as before.

The corrections are,

$$\left|\widetilde{\Psi}^{(0)}(x,t)\right\rangle = |\Psi(x,0)\rangle \tag{30}$$

$$\left|\widetilde{\Psi}^{(1)}(x,t)\right\rangle = \int_0^t \frac{\widetilde{\delta H}(t')}{i\hbar}\,|\Psi(x,0)\rangle\ dt' \tag{31}$$

$$\left|\widetilde{\Psi}^{(2)}(x,t)\right\rangle = \int_0^t dt'\ \frac{\widetilde{\delta H}(t')}{i\hbar}\int_0^{t'} dt''\ \frac{\widetilde{\delta H}(t'')}{i\hbar}\,|\Psi(x,0)\rangle \tag{32}$$

More generally, the n-th order equation takes the form,

$$i\hbar\partial_t\left|\widetilde{\Psi}^{(n)}(x,t)\right\rangle = \widetilde{\delta H}\left|\widetilde{\Psi}^{(n-1)}(x,t)\right\rangle \tag{33}$$

which means the solution can be written in the form,

$$\left|\widetilde{\Psi}^{(n)}(x,t)\right\rangle = \int_0^t \frac{\widetilde{\delta H}(t')}{i\hbar}\left|\widetilde{\Psi}^{(n-1)}(x,t)\right\rangle\ dt' \tag{34}$$

and the last integral can be expanded as an n times iterated integral based on the form of the (n-1)th order solution. In the end, we can apply the $H^{(0)}$ time evolution operator to get back the state $\Psi(x,t)$ we were originally working with.

### 2.3.3 Transition amplitudes

A direct use of the calculations we've performed previously is to figure out the probability of detecting the system in a specific state at time $t$, knowing the initial state we started with at $t=0$. For this purpose, the IP state ket can be used directly.

Let $|\Psi(x,0)\rangle = |n\rangle$ evolve under the time dependent Hamiltonian $H(t)$. The probability for it to be in state $|m\rangle$ at time $t$ is,

$$P_{m\leftarrow n}(t) = |\,\langle m|\Psi(x,t)\rangle\,|^2 = \left|\,\langle m|\,e^{-\frac{iH^{(0)}t}{\hbar}}\left|\widetilde{\Psi}(x,t)\right\rangle\right|^2 = \left|\left\langle m\middle|\widetilde{\Psi}(x,t)\right\rangle\right|^2 \tag{35}$$

Since the solution for the IP state ket is a series, we can calculate the probability of transition up to an arbitrarily high order. The first term in this series is $|\Psi(x,0)\rangle = |n\rangle$, the inner product of which with $|m\rangle$ vanishes. So,

$$P_{m\leftarrow n}(t) = \left|\left\langle m\middle|\widetilde{\Psi}^{(1)}(x,t)\right\rangle + \left\langle m\middle|\widetilde{\Psi}^{(2)}(x,t)\right\rangle + \dots\right|^2 \tag{36}$$

To first order,

$$P^{(1)}_{m\leftarrow n}(t) = \left|\int_0^t \frac{\langle m|\,\widetilde{\delta H}(t')\,|\Psi(x,0)\rangle}{i\hbar}\ dt'\right|^2 = \left|\int_0^t \frac{\langle m|\,\widetilde{\delta H}(t')\,|n\rangle}{i\hbar}\ dt'\right|^2 \tag{37}$$

Then using Eq. 17,

$$P^{(1)}_{m\leftarrow n}(t) = \left|\int_0^t e^{i\omega_{mn}t'}\,\frac{\delta H_{mn}(t')}{i\hbar}\ dt'\right|^2 \tag{38}$$

A wonderful corollary of this result is,

**Sidebar 2.4.** $P^{(1)}_{m\leftarrow n}(t) = P^{(1)}_{n\leftarrow m}(t)$
The reason behind this is

We can also derive the corrections in terms of the coefficients $c_n(t)$ for the IP state ket. This would require expressing each correction state ket as a linear combination of the basis states of $H^{(0)}$ and forming a series in $\lambda$ for the coefficient associated to say, state $n$.

The first order correction takes the form,

$$c^{(1)}_m(t) = \sum_n \int_0^t dt' e^{i\omega_{mn}t'} \frac{\delta H_{mn}(t')}{i\hbar} c_n(0) \tag{39}$$

and we know $c_n(0)$ for all $n$ since we know $|\Psi(x,0)\rangle$.

By definition,

$$P_{m\leftarrow n}(t) = \left| \left\langle m \middle| \widetilde{\Psi}(x,t) \right\rangle \right|^2 = |c_m(t)|^2 \tag{40}$$

and up to first order in perturbation theory,

$$P_{m\leftarrow n}(t) \approx |c^{(0)}_m(t) + c^{(1)}_m(t)|^2 = |c_m(0) + c^{(1)}_m(t)|^2 \tag{41}$$

$$= |c_m(0)|^2 + c_m(0)^* c^{(1)}_m(t) + c^{(1)}_m(t)^* c_m(0) + \mathcal{O}(\delta H^2) \tag{42}$$

### 2.3.4 Types of Perturbations and Fermi's Golden Rule

(Constant and harmonic perturbations)

# 3 Relativistic formulation of Electrodynamics

Our notation is as follows - we will use a 3-vector, denoted by $\vec{v}$, for spatial coordinates and 4-vectors for space-time (contravariant, for now) coordinates, denoted by $A^\mu$ where $\mu = 0, 1, 2, 3$. For instance, the 4-momentum $p^\mu$ of a particle will have the components $p^0 = E/c, p^1 = p_x, p^2 = p_y, p^3 = p_z$.

Let $\vec{u} = \frac{d\vec{l}}{dt}$ denote the ordinary velocity of a particle (from a far away, external frame). So $d\vec{l}$ and $dt$ are measured in that frame. Then the proper velocity of a particle is defined as,

$$\vec{\eta} = \frac{d\vec{l}}{d\tau} \tag{43}$$

Particularly,

$$\eta^0 = \frac{dx^0}{d\tau} = c \cdot \gamma = \frac{c}{\sqrt{1 - \frac{u^2}{c^2}}} \tag{44}$$

If the Lorentz transformation $\Lambda$ takes us from frame $S$ to frame $\bar{S}$, then the proper velocities in those two frames are related by,

$$\bar{\eta}^\mu = \Lambda^\mu_\nu \eta^\nu \tag{45}$$

This is an example of the transformation rule for 4-vector under Lorentz transforms (and in the way a 3-vector is defined by how it changes under spatial transformations, we can take this to be the defining rule for 4-vectors).

So, $\bar{\eta}^0 = \gamma(\eta^0 - \beta\eta^1)$ and so on for the transformation (which is an x-boost by speed $v = \beta c$),

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{46}$$

The transformation rules for ordinary velocities are a bit more complicated since we have to change both the infinitesimal displacement and infinitesimal time when we shift to a moving frame. They are (for a frame moving with speed $v$ in the positive x-axis direction),

$$\bar{u}_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} \tag{47}$$

$$\bar{u}_y = \frac{u_y}{\gamma\left(1 - \frac{u_x v}{c^2}\right)} \tag{48}$$

$$\bar{u}_x = \frac{u_z}{\gamma\left(1 - \frac{u_x v}{c^2}\right)} \tag{49}$$

The relativistic momentum $\vec{p}$ is defined using proper velocity since if we use ordinary ve-

locities, the law of conservation of momentum will be inconsistent with the principles of relativity.

**Fact check 3.1.** What if we defined relativistic momentum using ordinary velocities and not proper velocities?

So [4],

$$\vec{p} = m\vec{\eta} = \gamma m\vec{u} \tag{50}$$

The four momentum is constructed using the relativistic momentum for the last three coordinates, while the zero-th component of $p^\mu$ is,

$$p^\mu = m\eta^0 = \gamma mc \tag{51}$$

Identifying $E = p^0 c$ as the relativistic energy,

$$E = \gamma mc^2 \tag{52}$$

The rest energy (put $u = 0$ so that $\gamma = 1$) is $E_{rest} = mc^2$. Then the kinetic energy is $E_{KE} = E - E_{rest}$. If $u << c$, we can Taylor expand gamma (as a function of $u^2/c^2$, around 0) to get,

$$E_{KE} = \frac{1}{2}mu^2 + \frac{3}{8}\frac{mu^4}{c^2} + \dots \tag{53}$$

which agrees with the classical KE up to first order. This reaffirms our faith that SR should contain in it the theory of classical kinematics in the slow velocities limit.

The transformation rules for electric and magnetic fields (under an x-boost of speed $+v$) are,

$$\begin{aligned} \bar{E}_x &= E_x & \bar{B}_x &= B_x \\ \bar{E}_y &= \gamma(E_y - vB_z) & \bar{B}_y &= \gamma\Big(B_y + \frac{v}{c^2}E_z\Big) \\ \bar{E}_z &= \gamma(E_z + vB_y) & \bar{B}_z &= \gamma\Big(B_z - \frac{v}{c^2}E_y\Big) \end{aligned} \tag{54}$$

Griffiths uses an example of a parallel plate capacitor to derive the formulae for fields perpendicular to direction of motion (x direction). The same setup is used to get the rule for $E_x$, while to get the rule for $B_x$ (which is absent if you orient capacitor in y-z plane), a long solenoid aligned parallel to the x-axis is used.

While the derivation of these rules using specific setups is illustrative, especially when you haven't seen it before, you do have to make the leap of faith that they generalize to any configuration of charges/currents and thus arbitrary E and B fields. It would be nice to add a more general derivation here.

[4] Here, $\gamma = 1/\sqrt{1 - u^2/c^2}$

The electric and magnetic fields at a point in the new inertial frame are related in a simple manner for some special cases. If $\vec{B} = 0$ at a point in the original frame, then in the new frame (at that point),

$$\bar{\vec{B}} = -\frac{1}{c^2}(\vec{v} \times \bar{\vec{E}}) \tag{55}$$

Similarly, if $\vec{E} = 0$ at a point in the original frame, then in the new frame (at that point),

$$\bar{\vec{E}} = \vec{v} \times \bar{\vec{B}} \tag{56}$$

# 4 The Electromagnetic field tensor and a new way of looking at E&M

A very simple example of a moving point charge getting attracted to/repelled by a current carrying wire tells us that electric fields and magnetic fields are in fact, entwined with each other. One person's E-field can be another's B-field, something that has been established through many more such experiments.

We, as humans, crave elegance in our theories. We eventually do have to let go of this aspect as a requirement, since truth and accuracy take precedence in determining the future of a theory (as they should) and whether it should be pursue further. Now, with all this symmetry and intermingling, it would be nice (if not immensely satisfying) if E-fields and B-fields descended from one thing - something much more general - like a common ancestor. Thankfully, that is the case.

The fact that a Lorentz transformation mixes the components of E and B fields together eliminates the possibility of them being individual 4-vectors. Turns out, the object combining these two fields is an asymmetric, second-rank tensor.

As a definition for tensors, we can again impose certain rules on how they should behave. However, two Lorentz transformations are required to carry out this modification, instead of one as in the case of a 4-vector. Also, we can represent tensors as $2 \times 2$ matrices, thus labelling their components by the 2 indices $\mu, \nu$ [5]. So, a 2-tensor $t^{\mu\nu}$ is an object that follows the following transformation rule,

$$\bar{t}^{\mu\nu} = \Lambda^{\mu}_{\lambda}\Lambda^{\nu}_{\sigma}t^{\mu\nu} \tag{57}$$

A symmetric 2-tensor would thus have 10 independent components, while an anti-symmetric 2-tensor would have 6. Starting with a general anti-symmetric 2-tensor,

$$t^{\mu\nu} = \begin{pmatrix} 0 & t_{01} & t_{02} & t_{03} \\ -t_{01} & 0 & t_{12} & t_{13} \\ -t_{02} & -t_{12} & 0 & t_{23} \\ -t_{03} & -t_{13} & -t_{23} & 0 \end{pmatrix}$$

and then applying an x-boost, we can obtain the necessary transformation rules. Directly comparing them with those for the E and B fields found before, we construct the Electromagnetic Field Tensor,

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix} \tag{58}$$

An alternate comparison (switching what transformation rule of E/B fields is compared with

[5] I haven't yet explored the math behind this, so maybe that's a digression for another day

the which general tensor rule) yields a similar object, $G^{\mu\nu}$ - the dual tensor [6]. Both these tensors give rise to the same rules for E and B fields.

We can define a current density 4-vector $J^\mu$ as,

$$J^\mu = (c\rho, \vec{J}) \tag{59}$$

where $\rho$ is the charge density and $\vec{J}$ is the usual 3-D current density. Similarly, define the 4-vector potential $A^\mu$ as,

$$A^\mu = (V/c, \vec{A}) \tag{60}$$

where $V$ is the scalar electric potential and $\vec{A}$ is the 3-D magnetic vector potential.

We know that charge is locally conserved, which is expressed by the continuity equation:

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \tag{61}$$

For a fixed reference frame,

$$\nabla \cdot \vec{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = \sum_{i=1}^{3} \frac{\partial J^i}{\partial x^i}$$

and

$$\frac{\partial \rho}{\partial t} = \frac{\partial (c\rho)}{\partial (ct)} = \frac{\partial J^0}{\partial x^0} \tag{62}$$

which leads us to conclude (using 19),

$$\frac{\partial J^\mu}{\partial x^\mu} = 0 \tag{63}$$

This is where the notation really shines - it places time and space on an equal footing and tells us something deeper about this object $J^\mu$ that we've formed. Precisely, that it is divergence-less in 4D space-time.

Encouraged by this delightful expression, we will now express Maxwell's equations as 2 very elegant equations. And the objects that will show up - you guessed it - are the field tensor and the dual tensor.

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = \mu_0 J^\mu \tag{64}$$

$$\frac{\partial G^{\mu\nu}}{\partial x^\nu} = 0 \tag{65}$$

Choosing specific values for $\mu$ in the first equation would give rise to Gauss' law ($\mu = 0$) and Ampère's law with Maxwell's correction ($\mu = 1, 2, 3$ give the 3 vectorial components of the eq.). Doing the same for the second equation would give us the eq. exhibiting the

[6] It's the Hodge dual of $F^{\mu\nu}$

divergence-less nature of B-fields and Faraday's law.

We can also express the Minkowski force on a charge $K^\mu$ using the field tensor as follows,

$$K^\mu = q\eta^\nu F^{\mu\nu} \tag{66}$$

Evaluating the zeroth component of this equation gives us (with the three dimensional dot product),

$$K^0 = \frac{\gamma q}{c}(\vec{u} \cdot \vec{E}) = \frac{1}{c}(\vec{F}_E \cdot \vec{\eta}) \tag{67}$$

The field tensor can be expressed in terms of the 4-vector potential.

$$F^{\mu\nu} = \frac{\partial A^\mu}{\partial x_\nu} - \frac{\partial A^\nu}{\partial x_\mu} \tag{68}$$

Note that the derivatives now carry a subscript instead of superscript, so this would cost us a minus sign for the temporal component (w.r.t. our original convention).

# 5 Transverse and Longitudinal components of fields

For a vector field, the curl-free component is often called the longitudinal component ($||$) and the divergence-free component is called the transverse component ($\perp$). So, for a field $\vec{A} = \vec{A}_{||} + \vec{A}_{\perp}$,

$$\boldsymbol{\nabla} \times \vec{A}_{||} = 0 \tag{69}$$

$$\boldsymbol{\nabla} \cdot \vec{A}_{\perp} = 0 \tag{70}$$

This separation be performed for any sufficiently smooth and rapidly decaying vector field, and is called the Helmholtz decomposition. This might be most commonly recognized from the usual prescription of the electric field as $\vec{E} = -\boldsymbol{\nabla}\Phi - \boldsymbol{\nabla} \times \vec{A}$, where the scalar potential is irrotational and the vector potential is divergence-free (in the Coulomb gauge).

# 6 Plasmas

A general definition of plasma is as follows: a quasi-neutral gas of charged particles that shows collective behaviour.

The term quasi-neutral refers to the fact that even though plasma is made up of free electrons and ions that are negatively and positively charged, respectively, their overall charge densities are equal in equilibrium (locally, $n_e \approx Z \cdot n_i$, where $n_x$ refers to number density and $Z$ is the charged state of ions).

Further, we say collective behaviour to point to the fact that local changes in equilibrium can have strong influences on far away regions. Simply put, you can give rise to macroscopic changes via making microscopic disturbances - such as setting up a net electric field (which affects the response of the plasma to externally applied fields). It also refers to the domination of large-scale effects over individual particle collisions.

With these as desired traits, we can start to find physical parameters that would help us classify a system to be an ideal plasma. Our expectation is that plasma should tend to evolve towards a quasi-neutral state after disturbances. For instance, if we insert a probe net positive charge into its bulk, we'd hope to find that the electron density has increased near this probe charge and as a consequence, the potential that this charge would have originally created is screened. Thus, it's affect should die off rapidly with distance.

If the electron-ion soup is in thermal equilibrium ($T_e = T_i = T$), the ion velocity is almost an order of 2 smaller than electron velocity. We thus make the assumption that the ion density remains roughly constant ($n_i \approx n_0$). As it turns out (this I have yet to verify), electron density follows a Boltzmann distribution here, which allows us to conclude that,

$$n_e = n_i e^{e\phi/k_b T} \tag{71}$$

Using Gauss' law,

$$\boldsymbol{\nabla}^2 \phi = -\frac{\rho}{\epsilon_0} = -\frac{e(n_0 - n_e)}{\epsilon_0} \tag{72}$$

With spherical symmetry (no angular dependence - emerging from the probe or plasma), the potential can be derived to be,

$$\phi = \frac{1}{4\pi\epsilon_0}\frac{e^{-r/\lambda_D}}{r} \tag{73}$$

where $\lambda_D = \sqrt{\frac{\epsilon_0 k_b T}{e^2 n_e}}$ is the Debye length and acts as a length scale for our potential. It depends on the temperature and density of the plasma.

If we find that there are many particles per Debye sphere,

$$N = n_e \frac{4\pi}{3}\lambda_D^3 >> 1 \tag{74}$$

the density is high enough that the "collective behaviour" dominates over individual particle interactions and the plasma behaves ideally. Stay tuned for more clarification on this. Equivalently, we can set,

$$g = \frac{1}{n_e \lambda_D^3} \tag{75}$$

and demand that $g << 1$ for a classical, ideal plasma. This can then be the parameter we were searching for.

With $N_D \leq 1$, the screening effects are much less pronounced, and collisions dominate dynamics (again, the exact meaning of this remains unclear).

## 6.1 Plasma oscillations

When we displace a group of electrons in a small area of the plasma slightly, the net charge density in the new position is negative, and the net density in the original position is positive with the same magnitude. So, using a slab/capacitor model for a sheet of electrons, we expect this sheet to be pulled back by the restoring electric field set up as a result of the displacement $\delta$. The generated $E$-field is $E = \frac{en_e\delta}{\epsilon_0}$ and the equation of motion for this sheet turns out to be,

$$\frac{d^2\delta}{dt^2} + \omega_P \delta = 0 \tag{76}$$

where

$$\omega_P = \sqrt{\frac{e^2 n_e}{\epsilon_0 m_e}} \tag{77}$$

is the electron plasma frequency. This can also be derived by looking at the time it takes for the plasma to adjust to a new charge inserted in its bulk - as an estimate, this would be the ratio of the electron thermal velocity and the Debye length [7]. So,

$$t_P \approx \frac{\lambda_D}{v_e} = \omega_P^{-1} \tag{78}$$

If electromagnetic radiation with period more than the response time of the plasma ($\omega < \omega_P$) is incident on it, it would be shielded out. On the flip side, $\omega > \omega_P$ would mean the radiation is able to pass through, but is refracted and diminishes in intensity.

## 6.2 Waves in plasmas

To analyse wave phenomena in plasmas, methods such as the two-fluids equations, magnetohydrodynamics, etc. are used.

[7] Electrons far away would have to move less than the Debye length but those close to where the charge was inserted would be displaced considerably. On average, we take the distance moved to conform to the emergent potential to be the Debye length since its a natural length scale of the problem.

### 6.2.1 Longitudinal (Langmuir) waves

Using the two-fluids equations (restricted to one dimension for propagation), invoking Gauss' law to get an expression for the $E$ field, linearizing the non-linear system of ODEs, performing some algebraic acrobatics, and then looking for plane wave solutions gives us the following dispersion relation,

$$\omega^2 = \omega_P^2 + 3k^2 v_{te}^2 \tag{79}$$

This is known as the Bohm–Gross dispersion relation. Up to first order in $v_{te}$, this relation is independent of $k$ ($\omega \approx \omega_P$).

### 6.2.2 Transverse waves

More math and electrodynamics yields the following dispersion relation for transverse waves in plasmas,

$$\omega^2 = \omega_P^2 + c^2 k^2 \tag{80}$$

With natural units (setting $c = 1$), we see this as indication that in the relativistic regime, transverse waves behave like particles with (effective) mass $m = \omega_P = (\omega^2 - k^2)^{1/2}$.

# 7 Density Matrices

# 8 Dispersion relations

# 9 Metric Tensor

From a coordinate-independent point of view, a metric tensor is a smooth, non-degenerate, symmetric tensor field - it assigns to each point of the manifold a symmetric bilinear form on its tangent space (with some specified regularity).

The components of a metric tensor take the form of a symmetric matrix in a coordinate basis. They transform covariantly under changes of coordinate system.

## References